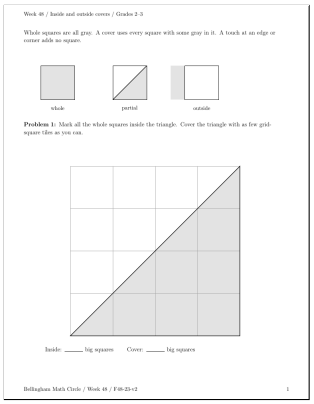
Week 48 / Inside and outside covers / Problem menu

for adults: pick, then print


Week 48 / Inside and outside covers / Grades 2–3

Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.

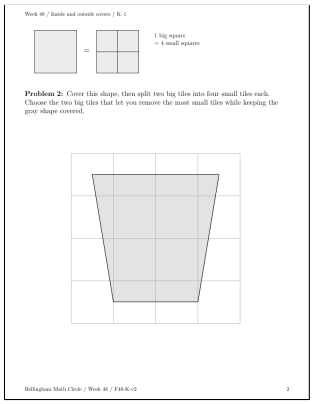
whole partial outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.

Inside: ____ big squares Cover: ____ big squares

Bellingham Math Circle / Week 48 / F48-23-v2 1

Inside and cover for the triangle
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route

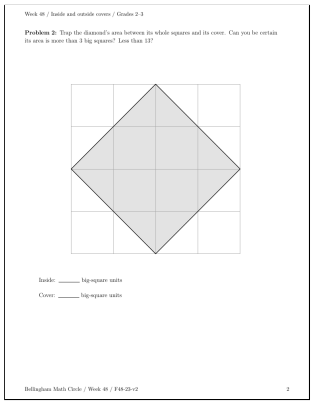

Week 48 / Inside and outside covers / K–1

= 1 big square = 4 small squares

Problem 2: Cover this shape, then split two big tiles into four small tiles each. Choose the two big tiles that let you remove the most small tiles while keeping the gray shape covered.

Bellingham Math Circle / Week 48 / F48-K-v2 2

Split two big tiles of the trapezoid cover
K–1 P2
return route


Week 48 / Inside and outside covers / Grades 2–3

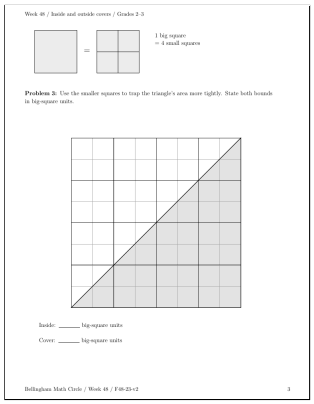
Problem 2: Trap the diamond's area between its whole squares and its cover. Can you be certain its area is more than 3 big squares? Less than 13?

Inside: ____ big-square units

Cover: ____ big-square units

Bellingham Math Circle / Week 48 / F48-23-v2 2

Trap the diamond's area
Grades 2–3 P2 · Grades 4–5 P2
first route


Week 48 / Inside and outside covers / Grades 2–3

= 1 big square = 4 small squares

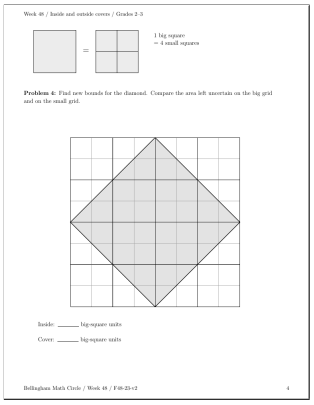
Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.

Inside: ____ big-square units

Cover: ____ big-square units

Bellingham Math Circle / Week 48 / F48-23-v2 3

The triangle with small squares
K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3
first route
wording differs by band


Week 48 / Inside and outside covers / Grades 2–3

= 1 big square = 4 small squares

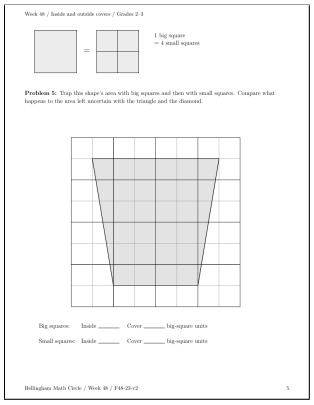
Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.

Inside: ____ big-square units

Cover: ____ big-square units

Bellingham Math Circle / Week 48 / F48-23-v2 4

The diamond with small squares
K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4
K-1 return; 2-3 first; 4-5 first route
wording differs by band


Week 48 / Inside and outside covers / Grades 2–3

= 1 big square = 4 small squares

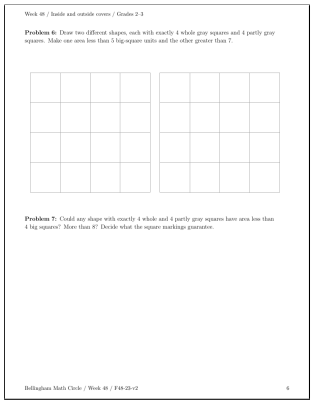
Problem 5: Trap this shape's area with big squares and then with small squares. Compare what happens to the area left uncertain with the triangle and the diamond.

Big squares: Inside ____ Cover ____ big-square units

Small squares: Inside ____ Cover ____ big-square units

Bellingham Math Circle / Week 48 / F48-23-v2 5

The trapezoid on both grids
Grades 2–3 P5
return route

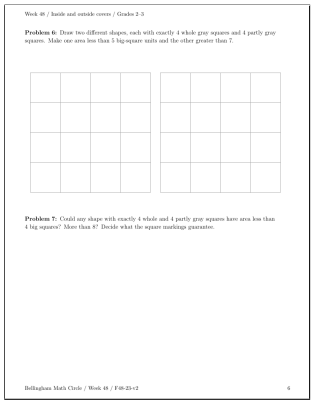

Week 48 / Inside and outside covers / Grades 2–3

Problem 6: Draw two different shapes, each with exactly 4 whole gray squares and 4 partly gray squares. Make one area less than 5 big-square units and the other greater than 7.

Problem 7: Could any shape with exactly 4 whole and 4 partly gray squares have area less than 4 big squares? More than 8? Decide what the square markings guarantee.

Bellingham Math Circle / Week 48 / F48-23-v2 6

Same markings, different areas
K–1 P5 · Grades 2–3 P6
return route
wording differs by band

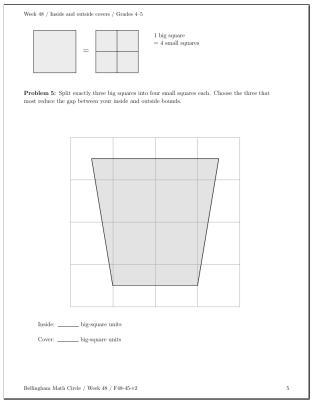

Week 48 / Inside and outside covers / Grades 2–3

Problem 6: Draw two different shapes, each with exactly 4 whole gray squares and 4 partly gray squares. Make one area less than 5 big-square units and the other greater than 7.

Problem 7: Could any shape with exactly 4 whole and 4 partly gray squares have area less than 4 big squares? More than 8? Decide what the square markings guarantee.

Bellingham Math Circle / Week 48 / F48-23-v2 6

What the markings guarantee
K–1 P6 · Grades 2–3 P7
return route
wording differs by band


Week 48 / Inside and outside covers / Grades 4–5

= 1 big square = 4 small squares

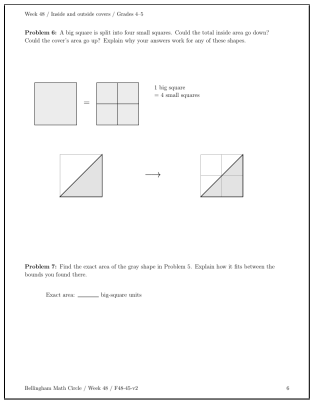
Problem 5: Split exactly three big squares into four small squares each. Choose the three that most reduce the gap between your inside and outside bounds.

Inside: ____ big-square units

Cover: ____ big-square units

Bellingham Math Circle / Week 48 / F48-45-v2 5

Spend three splits on the trapezoid
Grades 4–5 P5
return route


Week 48 / Inside and outside covers / Grades 4–5

Problem 6: A big square is split into four small squares. Could the total inside area go down? Could the cover's area go up? Explain why your answers work for any of these shapes.

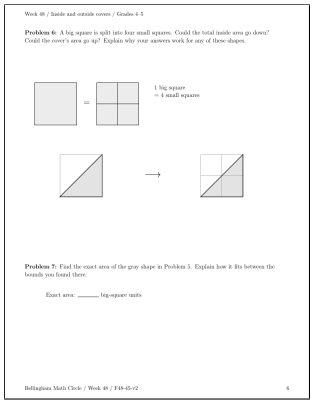
= 1 big square = 4 small squares

Problem 7: Find the exact area of the gray shape in Problem 5. Explain how it fits between the bounds you found there.

Exact area: ____ big-square units

Bellingham Math Circle / Week 48 / F48-45-v2 6

Can a split make a bound worse?
Grades 4–5 P6
return route


Week 48 / Inside and outside covers / Grades 4–5

Problem 6: A big square is split into four small squares. Could the total inside area go down? Could the cover's area go up? Explain why your answers work for any of these shapes.

= 1 big square = 4 small squares

Problem 7: Find the exact area of the gray shape in Problem 5. Explain how it fits between the bounds you found there.

Exact area: ____ big-square units

Bellingham Math Circle / Week 48 / F48-45-v2 6

The trapezoid's exact area
Grades 4–5 P7
return route